MILITARY SPECIFICATION

CONTROL-MONITOR DCU-116/A

1. SCOPE. This specification covers the requirements for one type of control-monitor designated as the DCU-116/A for multi-station nuclear bomb control and monitor functions in high performance aircraft.

2. APPLICABLE DOCUMENTS. The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

Military

MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-I-8500	Interchangeability and Replaceability, Physical, of Component Parts for Aircraft
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery Of

Standards

MIL-STD-16	Electric and Electronic Reference Designations
MIL-STD-100	Drawings, Administrative Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation Of

Drawings

66J13220	Control-Monitor DCU-116/A

(Copies of the above documents which are required by contractors in connection with specific procurement functions should be obtained from the Administrative Contracting Officer or as directed by the procuring activity.)

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3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Design and Construction. The control-monitor shall be designed and constructed in conformance to Drawing No. 66J13220 and subsidiary drawings.

3.2.1 Interchangeability. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable, the criteria of MIL-D-1000 shall apply and new drawings shall be prepared as required. Physical interchangeability shall be in accordance with MIL-I-8500.

3.2.2 Electrical Wiring. All insulated stranded electrical wire used to interconnect components shall be of 19 strand construction and meet the requirements of MIL-W-16878, Type E.

3.2.3 Finish. Finish of the Control-Monitor shall be in conformance to the engineering drawings.

3.2.4 Dielectric Strength. The electrical conducting parts shall be insulated from the case. The insulation shall be capable of withstanding repeated applications of at least one minute duration of voltages up to 1000 volts rms.

3.3 Performance. Satisfactory performance of the control-monitor shall consist of the capability to control the functioning of bomb circuits and monitor the condition of these circuits.

3.3.1 Reliability. The control-monitor shall have a Mean Time Between Failure (MTBF) rating of 350 cycles of operation under service conditions.

3.4 Insulation Resistance. The insulation resistance between the specified pins shall be measured using 500 $\pm$ 25 volts DC applied to each of the circuits. The current shall not exceed 100 microamperes or the insulation resistance between shall be at least 5.0 megohms.

3.5 Environmental Requirements. The control-monitor shall be capable of satisfactory performance during or after, as applicable, subjection to the following environmental conditions:

a. Low Temperature and Thermal Shock. The unit with test cable connected shall be exposed to a temperature of -65°F for at least three hours. Immediately upon completion of the above test, the unit shall be exposed to a temperature of 160°F for at least one hour.

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b. High Temperature and Thermal Shock. The unit with test cable connected shall be exposed to a temperature of 160°F for at least three hours with an operating voltage of 22 volts DC. Immediately upon completion of the above test, the unit shall be exposed to a temperature of -65°F for at least one hour.

c. Vibration from 10 to 2000 cps sinusodial at a level of 3g or total excursion of 0.1 inch amplitude whichever is the least severe.

d. Humidity such as encountered in tropical areas.

e. Shock incident to transportation, handling and service conditions.

3.6 Identification of Product. The control-monitor, parts and assemblies shall be marked for identification in accordance with MIL-STD-130.

3.7 Reference Designations. Reference designations used for marking within the unit shall be in accordance with MIL-STD-16. Markings shall not adversely affect the leakage path between conductors, or any other factor of performance. Reference symbols shall be employed to identify each part for its particular circuit application. The same reference symbol shall be used for identifying a particular part on all types of data, including drawings, diagrams, and spare parts lists prepared for the equipment. When practicable, markings shall be placed on the supporting structure adjacent to each replaceable part in such a position as to be readily reviewed.

3.8 Government Loaned Property. When so stated in the contract, the following item will be furnished the contractor, upon his request, as Government loaned property for the purpose of testing the control-monitor:

ITEM NUMBER	DESCRIPTION OF ITEM	QUANTITY
1	PT 1239 Test Set with U-3319 Test Fixture or equal	1

3.9 Workmanship. The control-monitor shall be fabricated and finished in a manner such that the criteria of appearance, fit and adherence to specified tolerances shall be observed. These criteria shall apply to all nonfunctioning parts as well as to functioning parts. Particular attention shall be given to marking of parts and assemblies, plating, machine screw assemblage, and freedom of parts from burrs and sharp edges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests. Inspection and testing of the control-monitor shall be classified as follows:

a. Preproduction Tests

b. Acceptance Tests

4.2 Preproduction Testing

4.2.1 Preproduction Test Sample. The preproduction test units shall be representative of the production equipment. They shall be tested by a laboratory designated by the procuring activity, or when so stated in the contract, at the contractor's facility under the supervision of the procuring activity.

4.2.2 Preproduction Report. In the event that preproduction testing is performed at the contractor's facility the contractor shall, at the completion of preproduction test, prepare a preproduction test report. The preproduction test report shall be prepared in accordance with MIL-STD-831 and three complete copies shall be furnished to the procuring activity.

4.2.3 Preproduction Tests. Preproduction tests shall consist of all tests described in "4.5 Test Methods" of the specification.

4.3 Acceptance Tests. Acceptance tests shall consist of the following:

a. Individual Tests

b. Sampling Tests

4.3.1 Individual Tests. Each control-monitor shall be subjected to the following individual tests:

a. Examination of product.

b. Conformance to drawings.

c. Functional tests and Lamp tests.

4.3.2 Sampling Tests. Sampling tests shall consist of the following and shall be performed in the order shown:

a. Low Temperature and Thermal Shock test

b. High Temperature and Thermal Shock test

c. Vibration test

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4.3.2.1 Sampling Plan. Control-Monitors to be subjected to the sampling tests shall be selected at random from production items as follows:

a. First 100 produced, 1 sample from each lot of 10.

b. Each additional 1000 produced, 1 sample from each lot of 100.

4.3.2.2 Rejection and Retest. When a sample item fails to satisfy the specified tests, no item still on hand or later produced shall be accepted until the extent and cause of failure is determined and resolved.

4.3.2.3 Individual Test May Continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of items on hand produced later shall not be made until it is determined that the items meet all requirements of this specification.

4.3.3 Defects In Items Already Accepted. The investigation of a test failure could indicate that defects exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test Conditions. Unless otherwise specified, tests shall be performed at room atmospheric conditions, and the unit shall be at room temperature when each test is started.

4.5 Test Methods

4.5.1 Examination of Product. Each control-monitor shall be examined visually to determine compliance with all requirements of this specification for which tests are not specified. Failure to meet these requirements shall be cause for rejection.

4.5.2 Conformance to Drawings. Each control-monitor shall be inspected to determine conformance to the engineering drawings, subsidiary drawings and all specifications and standards referenced therein. Failure to conform to these criteria shall be cause for rejection.

4.5.3 Insulation Resistance Tests. The resistance between the specified pins shall be measured using 500 $\pm$ 25 volts DC applied to each of the circuits specified below. The voltage shall be applied for at least 2 but not more than 60 seconds. The current shall not exceed 100 microamperes or the insulation resistance shall be at least 5.0 megohms. Prior to performing this test, remove all lamps in the DCU-116/A (see figure 1) from their respective receptacles, rotate the

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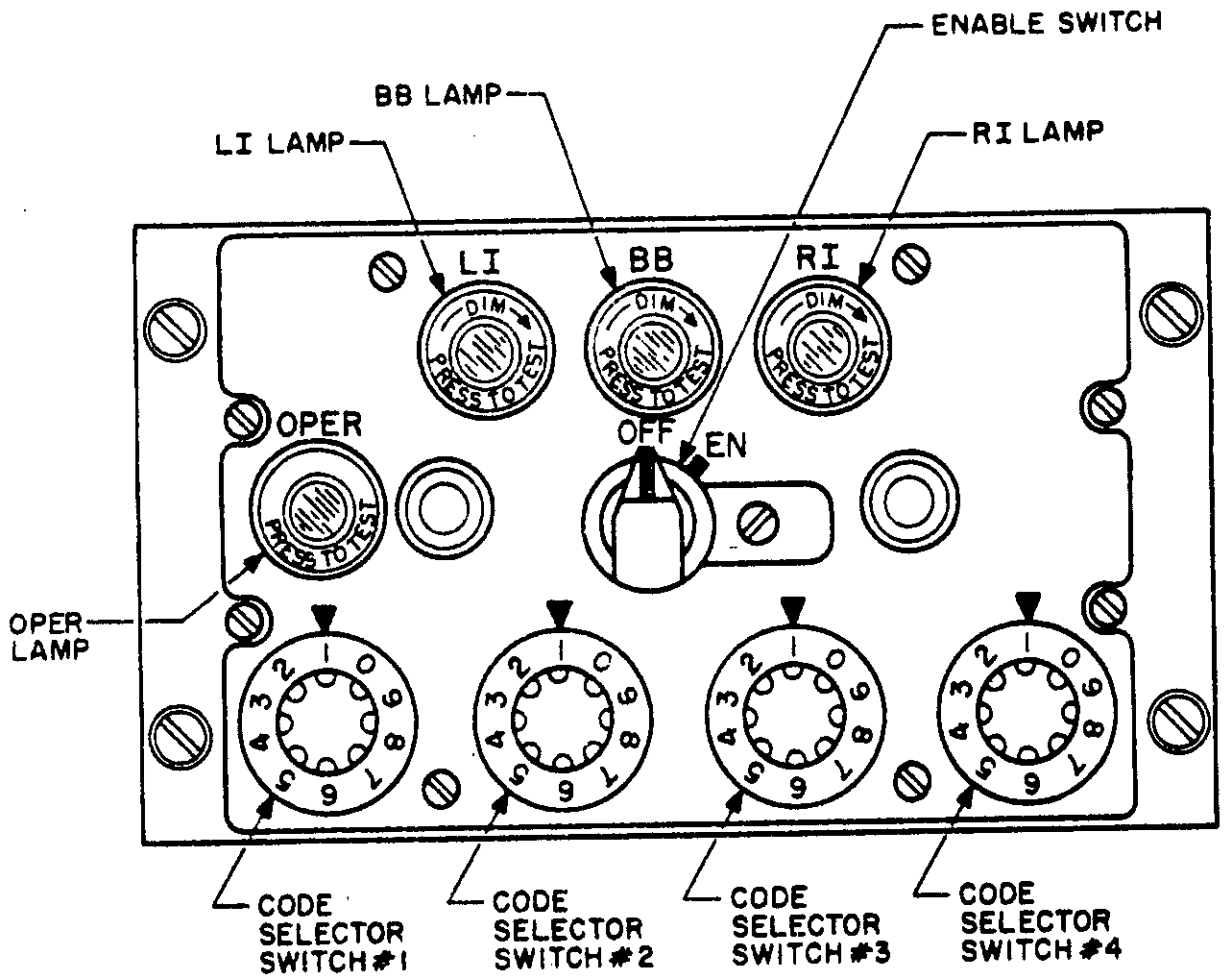


Figure 1 - DCU-116/A PANEL LAYOUT

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Enable switch to the EN position, and apply 28 volts DC between pin e and pin M or N (pin e positive) for 60 seconds minimum. While the voltage is still applied, rotate the Enable switch to the OFF position and allow 6 seconds for the unit to reset. Unless otherwise specified, the positional settings of the Code Selector switches are optional during this test.

From Pin	To
e	All other pins connected to pin M
Z	All other pins connected to pin M
F	All other pins connected to pin M
G	All other pins connected to pin M
H	All other pins connected to pin M
J	All other pins connected to pin M
K	All other pins connected to pin M
A	All other pins connected to pin M
B	All other pins connected to pin M
C	All other pins connected to pin M
D	All other pins connected to pin M
E	All other pins connected to pin M
L	All other pins connected to pin M
S	All other pins connected to pin M
T	All other pins connected to pin M
V	All other pins connected to pin M
Y	All other pins connected to pin M
a	All other pins connected to pin M
b	All other pins connected to pin M
c	All other pins connected to pin M
X	All other pins, except W, connected to pin M
U	All other pins, except d, connected to pin M

4.5.4 Functional Tests. The functional test on the permissive action link portion of the DCU-116/A shall be performed as follows:

a. Apply 28 volts DC between pin e and pins M and N (pin e positive).

b. Rotate the Enable switch (see Figure 1) to the EN position. Allow 60 seconds minimum for the unit to reach the end-point position.

c. Rotate the Enable switch to the OFF position. Allow 6 seconds minimum for the unit to reset.

d. Rotate the four Code Selector switches (see Figure 1) to the 0000 position.

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e. Rotate the Enable switch to the EN position and allow the unit to reach the end-point position.

f. Repeat steps c, d, and e using the following codes set with the Code Selector switches. The code operations shall be performed in the order specified.

Code	Code Selector switch positions (see Figure 1)			
	Switch No. 1	Switch No. 2	Switch No. 3	Switch No. 4
2	8	8	8	8
3	2	2	2	2
4	1	1	1	1
5	4	4	4	4
6	6	6	6	6
7	5	5	5	5
8	7	7	7	7
9	3	3	3	3
10	9	9	9	9

g. The unit shall meet the following requirements:

(1) OPER Lamp. The OPER lamp shall remain on during the entire operating time of the DCU-116/A.

(2) Pulse Order. The order of pulses shall be as follows:

	Preliminary Pulses	Prefix Code	Code								Suffix Code		
Index Pulse	-	12	-	1	-	1	-	1	-	1	-	1	→ Time
Code Pulse	0 to 10*		C_1	-	C_2	-	C_3	-	C_4	-	10	-	

*Due to motor coasting, the number of code pulses produced can vary from 0 to 10.

(3) Code. The code is set by means of the four Code Selector switches. C_1 shall represent the number of code pulses that is produced as a result of the setting on Code Selector switch No. 2, C_2 the number of code pulses that is produced as a result of the setting on Code Selector switch No. 4, C_3 the number of code pulses that is produced as a result of the setting on Code Selector switch No. 1, and C_4 the number of code pulses that is produced as a result of the setting on Code Selector switch No. 3. The relation between the setting on any Code Selector switch and the number of code pulses produced is given below.

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Setting of any Code Selector Switch	Number of pulses produced as a result of setting on Code Selector Switch No. 2 (C_1)	Number of pulses produced as a result of setting on Code Selector Switch No. 4 (C_2)	Number of pulses produced as a result of setting on Code Selector Switch No. 1 (C_3)	Number of pulses produced as a result of setting on Code Selector Switch No. 3 (C_4)
0	8	2	1	3
1	7	6	4	2
2	3	8	9	4
3	5	1	2	10
4	6	7	8	9
5	9	4	10	8
6	2	5	3	7
7	10	9	6	1
8	4	10	7	6
9	1	3	5	5

(4) Code Time. The total time for the unit to operate from reset to the end-point position shall not exceed 40 seconds. The tolerance for the time measurement shall be $\pm$ 5 percent.

(5) "On" Pulse Time. The DCU-116/A is designed to produce an "On" time pulse of 125 milliseconds minimum duration. The "on" time duration shall be monitored for all pulses (except preliminary pulses). Actual time measurements shall be performed on a sample basis. The tolerance for the time measurements shall be as follows:

(a) Actual Time Measurements. The test device shall accept any unit producing an "on" time pulse of 125 milliseconds or greater. The test device must reject any unit producing an "on" time pulse of less than 122 milliseconds. Units producing "on" time pulses ranging from 122 to 125 milliseconds may be accepted or rejected depending on the tolerance condition of the test device.

(b) Pulse Monitoring. The test device shall accept any unit producing an "on" time pulse of 125 milliseconds or greater. The test device must reject any unit producing an "on" time pulse of less than 112 milliseconds. Units producing "on" time pulses ranging from 112 to 125 milliseconds may be accepted or rejected depending on the tolerance condition of the test device.

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During the operation of code 5555, the "on" time duration of the following pulses shall be measured.

Code Selector Switch Pulse	Code Pulse
C_1	8th
C_2	2nd
C_3	10th
C_4	6th
Suffix	4th

During the operation of code 8888, the "on" time duration of the following pulses shall be measured.

Code Selector Switch Pulse	Code Pulse
C_1	3rd
C_2	9th
C_3	5th
C_4	1st
Suffix	7th

(6) "Off" Pulse Time. The DCU-116/A is designed to produce an "off" time duration between any two successive pulses of 150 milliseconds or greater. The "off" time duration between all successive pulses shall be monitored (except during preliminary pulses). Actual time measurements shall be performed on a sample basis. The tolerance for the time measurements shall be as follows:

(a) Actual Time Measurements. The test device shall accept any unit producing an "off" time duration of 150 milliseconds or greater. The test device must reject any unit producing an "off" time duration of less than 147 milliseconds. Units producing "off" time durations ranging from 147 to 150 milliseconds may be accepted or rejected depending on the tolerance condition of the test device.

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(b) Pulse Monitoring. The test device shall accept any unit producing an "off" time duration of 150 milliseconds or greater. The test device must reject any unit producing an "off" time duration of less than 135 milliseconds. Units producing "off" time durations ranging from 135 to 150 milliseconds may be accepted or rejected depending on the tolerance condition of the test device.

During the operation of code 5555, the "off" time duration between the following pulses shall be measured.

Code Selector Switch Pulse	Between Code Pulses
C_1	8th and 9th
C_2	2nd and 3rd
C_3	10th and the following index
C_4	6th and 7th
Suffix	4th and 5th

During the operation of code 8888, the "off" time duration between the following pulses shall be measured.

Code Selector Switch Pulse	Between Code pulses
C_1	3rd and 4th
C_2	9th and 10th
C_3	5th and 6th
C_4	1st and 2nd
Suffix	7th and 8th

4.5.5 LAMP. For this test, all lamps shall be installed in their respective receptacles and the test shall be performed as follows:

a. Apply 28 volts DC between pin e and pins M and N (pin e positive) and between pin Z and pins M and N (pin Z positive). The two panel lights shall come on.

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b. Depress the lens of the L1 lamp momentarily to actuate the press-to-test feature of the lamp.

CAUTION: Do not tighten the cap assembly of the press-to-test lamp excessively since overcompression of rubber results in excessive operating force.

c. Depress the lens of the BB/C lamp momentarily to actuate the press-to-test feature of the lamp.

CAUTION: Do not tighten the cap assembly of the press-to-test lamp excessively since overcompression of rubber results in excessive operating force.

d. Depress the lens of the R1 lamp momentarily to actuate the press-to-test feature of the lamp.

CAUTION: Do not tighten the cap assembly of the press-to-test lamp excessively since overcompression of rubber results in excessive operating force.

e. Depress the lens of the OPER lamp momentarily to actuate the press-to-test feature of the lamp.

CAUTION: Do not tighten the cap assembly of the press-to-test lamp excessively since overcompression of rubber results in excessive operating force.

f. Connect pin G to pins M and N. The L1 light shall come on.

g. Connect pin H to pins M and N. The BB/C light shall come on.

h. Connect pin J to pins M and N. The R1 light shall come on.

4.5.6 ENVIRONMENTAL TESTS. During or after subjection to the environmental tests, as specified in the test, the control-monitor shall be subjected to the functional tests. Failure to satisfy the requirements of the functional tests shall be considered as failure to pass the environmental test and cause for rejection.

4.5.6.1 High Temperature and Thermal Shock Test. The control-monitor shall be subjected to a temperature of +160°F for at least three hours. The functional tests, insulation resistance test and lamp tests in that order shall be performed at this specific temperature with an operating voltage of 22 volts DC. Immediately upon completion at the above tests, the control-monitor shall be exposed to a temperature of -65°F for at least one hour. Upon completion of the exposure and while at -65°F the control-monitor shall be subjected to the functional tests and insulation resistance

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test to the functional tests with an operating voltage of 33 volts DC. Visual inspection shall reveal no damage that might ultimately result in malfunction.

Note: The test chamber may be opened to perform the lamp test.

4.5.6.2 Low Temperature and Thermal Shock Test. The control-monitor shall be subjected to a temperature of -65°F for at least three hours. The insulation resistance test, functional tests and lamp tests in that order, will be performed at this specific temperature with an operating voltage at 33 volts DC. Immediately upon completion of the above test, the control-monitor shall be exposed to a temperature of 160°F for at least 1 hour. Upon completion of temperature exposure and while at the 160°F the control-monitor shall be subjected to the insulation resistance tests and functional tests, with an operating voltage of 22 volts DC. Visual inspection shall reveal no damage that might ultimately result in malfunction.

Note: The test chamber may be opened to perform the lamp tests. An elasped time of 2 minutes minimum between code cycles shall be allowed for temperature stabilization.

4.5.6.3 Humidity Test. The humidity test shall be performed in accordance with MIL-STD-810, Method 507.1, Procedure 1. Within one hour after the conclusion of the test, the insulation test, functional test and lamp test, in that order, shall be performed.

4.5.6.4 Shock Test. The shock test shall be performed in accordance with MIL-STD-810, Method 516.1, Procedure 1. At conclusion of the test, the control-monitor shall satisfy the requirements of the functional tests and lamp tests.

4.5.6.5 Vibration Test. Vibration testing of the control-monitor shall be performed in accordance with MIL-STD-810, Method 514.1, Test Curve C. At conclusion of test, equipment operation shall consist of the functional tests and lamp tests. Opening or closing of any circuit due to vibration or failure to satisfy the requirements of the functional tests shall be cause for rejection.

4.5.7 RELIABILITY TEST. The control-monitor shall be subjected to a reliability test to determine mean time between failure (MTBF) and expected service life. The reliability test shall consist of performance of the functional test repeated 250 times. Each performance of the functional test shall be considered one cycle. Conditions for this test shall be:

a. Vibration - .02 inch double amplitude at 25 to 35 cps for 10 minutes of each hour of the test.

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b. On-Off cycling - On for 3 hours and 10 minutes and off for 50 minutes of each four hour period of testing.

c. Temperature - $25^{\circ}C \pm 5^{\circ}C$

d. Altitude - 0 to 5000 feet above mean sea level.

e. Humidity - 90 percent or less.

f. A.C. voltage 115-400 cps nominal.

g. D.C. voltage.

Failure of any type prior to 175 cycles shall be cause for rejection. Failure after 175 cycles and before 350 cycles requiring adjustment by means other than nominal controls shall be cause for rejection.

5. PREPARATION FOR DELIVERY. Preparation of the control-monitor for delivery shall be in accordance with MIL-E-17555.

5.1 MARKING. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

CONTROL-MONITOR DCU-116/A

6. NOTES

6.1 INTENDED USE. The Control-Monitor DCU-116/A is intended for use in monitoring and control of weapons carried in and delivered by high performance aircraft.

6.2 ORDERING DATA. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Conditions under which Government loaned property will be made available to the contractor.

c. Whether preproduction tests will be performed by the procuring activity.

d. Levels of preservation, packaging and packing.

6.2.1 PREPRODUCTION SAMPLES. It is expected that the contract or purchase order will specify that a minimum of three units representative of the production control-monitor will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order should specify the point of inspection for these tests.

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6.3 PRODUCTION DELIVERY. No production control-monitor should be delivered prior to full approval of the preproduction samples.

6.4 DEFINITIONS

a. Preproduction Tests. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. Acceptance Tests. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

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